THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A hemispherical depression of diameter 42 cm, the largest possible, is scooped out of the top face of a cubical block of edge 42 cm. The volume of the remaining block is: 1

(a) 54684 cm³ **(b)** 74088 cm³ **(c)** 19404 cm³ **(d)** 35280 cm³

2. A solid metallic sphere of radius 9 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. The length of the wire is: 1

(a) 486 cm **(b)** 121.5 cm **(c)** 729 cm **(d)** 243 cm

3. A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 12 cm, and no water spills. The rise in the water level is: 1

(a) 2 cm **(b)** 0.5 cm **(c)** 4 cm **(d)** 1 cm

4. A capsule is a cylinder of radius 14 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 2112 cm² **(b)** 3344 cm² **(c)** 4576 cm² **(d)** 6688 cm²

5. A toy is a cone of base radius 28 cm and height 96 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 13728 cm² **(b)** 8800 cm² **(c)** 13376 cm² **(d)** 16192 cm²

6. A solid metal cuboid 24 cm by 18 cm by 15 cm is melted and recast into cubes of edge 3 cm. The number of cubes obtained is: 1

(a) 720 **(b)** 240 **(c)** 239 **(d)** 480

7. A tent is a cylinder of radius 7 m and height 8 m surmounted by a cone of the same radius and height 24 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 352 m² **(b)** 550 m² **(c)** 902 m² **(d)** 880 m²

8. A hemispherical bowl has outer radius 12 cm and inner radius 6 cm. The volume of material used to make the bowl is: 1

(a) 3168 cm³ **(b)** 1584 cm³ **(c)** 6336 cm³ **(d)** 9504 cm³

9. A hemispherical depression of diameter 63 cm, the largest possible, is scooped out of the top face of a cubical block of edge 63 cm. The volume of the remaining block is: 1

(a) 119070 cm³ **(b)** 65488.5 cm³ **(c)** 250047 cm³ **(d)** 184558.5 cm³

10. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 4 cm. The length of the wire is: 1

(a) 18 cm **(b)** 36 cm **(c)** 9 cm **(d)** 72 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid metallic cylinder of base radius 6 cm and height 72 cm is melted and recast into solid spheres of radius 2 cm. Find the number of spheres obtained. 2

12. A solid is a cube of edge 49 cm surmounted by a hemisphere of diameter 49 cm (the hemisphere rests on a face of the cube). Find the total surface area of the solid. 2

13. A solid metallic cone of base radius 12 cm and height 48 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

14. A solid metallic cylinder of base radius 9 cm and height 32 cm is melted and recast into solid spheres of radius 3 cm. Find the number of spheres obtained. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A capsule is a cylinder of radius 21 cm and length 12 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

16. A toy is a cone of base radius 10.5 cm and height 7 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

17. A wooden article is made by scooping out a hemisphere of radius 7 cm from each end of a solid cylinder of height 12 cm and base radius 7 cm. Find the total surface area of the article. 3

OR

A capsule is a cylinder of radius 21 cm and length 24 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

18. A toy is a cone of base radius 10.5 cm and height 35 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

OR

A wooden article is made by scooping out a hemisphere of radius 14 cm from each end of a solid cylinder of height 15 cm and base radius 14 cm. Find the total surface area of the article. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A conical vessel of radius 6 cm and height 12 cm is full of water. The water is poured into a cylindrical vessel of base radius 6 cm. Find the height of the water in the cylindrical vessel. 5

OR

Water from a full cylindrical tank of radius 7 cm and height 12 cm is poured into an empty rectangular tank of base 42 cm by 11 cm. Find the height of water in the rectangular tank. 5

20. A conical vessel of radius 6 cm and height 18 cm is full of water. The water is poured into a cylindrical vessel of base radius 6 cm. Find the height of the water in the cylindrical vessel. 5

OR

Water from a full cylindrical tank of radius 14 cm and height 10 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

END OF QUESTION PAPER — Total: 40 Marks